

AnalystLab Africa — Experience Lab Internship Programme
FinTrust Digital Bank — Week 1 Submission
Data Analytics Track
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TABLE OF CONTENTS

1. Business Understanding
2. Resources Review
3. Track-Specific Objectives
4. Success Criteria
5. Data Understanding and Profiling Report
6. Analytical Questions
7. KPI Definition Table
8. Dashboard Wireframe and Planning
9. Initial Analysis Plan
10. Relevant Risks and Future Dependencies

1. Business Understanding

1.1 What problem is FinTrust trying to solve?

FinTrust Digital Bank wants to use data, machine learning, and generative AI to better understand customer and transaction behaviour, support risk-related decision-making, improve operational processes, and strengthen digital customer support. From a Data Analytics perspective, the key opportunity is to transform the available customer and transaction data into meaningful business intelligence that can help FinTrust understand what is happening across its customers, transactions, channels, locations, transaction outcomes, and risk-review patterns.

1.2 Why is this problem important?

As a digital-first bank, FinTrust competes on customer experience and risk management rather than physical branch presence. Undetected risk patterns can lead to fraud losses and regulatory exposure, while poor operational visibility (failed transactions, channel reliability) directly damages customer trust and retention. The available data can provide insight into: Customer characteristics and segments, customer account types and engagement levels, transaction volumes and values, transaction trends over time, the performance and usage of different transaction channels, transaction success, failure, reversal, and pending outcomes, geographic patterns in transaction activity, risk-review patterns within the transaction data.

These insights can support business monitoring and provide useful evidence for subsequent work by other project tracks.

1.3 How can the Data Analytics track contribute?

The Data Analytics track will help turn raw customer and transaction records into structured business intelligence, profiling the available customer and transaction datasets for quality and structure, identifying & answering targeted analytical questions, defining KPIs that management can track over time, providing evidence-based insights that can support later cross-track collaboration, and designing a dashboard that surfaces these insights continuously.

1.4 What type of output can the Data Analytics track provide?

- A validated data profiling report (structure, quality, relationships)
- A set of answered analytical questions with supporting evidence
- A KPI definition table with real baseline figures
- Exploratory and descriptive analysis.
- A dashboard wireframe and, in later weeks, a working Power BI dashboard.

2. Resource Review

Three approved resources were used for the Data Analytics track this week:

Resource	Contents	How it was used	Limitations
FinTrust Customer Data	1,500 customer records: demographics, segment, account type, tenure, engagement score, income band, channel preference, and account status.	Profiled for structure and quality. Used for customer behaviour questions and KPI baselines.	Customer Name is for display only, not usable for modelling. Dataset is synthetic and represents a limited project period/context.
FinTrust Transaction Data	12,000 transaction records: type, amount, channel, device, location, international flag, status, risk flag.	Used to analyze transaction activity, value, channels, outcomes, locations, and risk-review patterns.	96 rows have missing Device Type and/or Location (intentional, per data dictionary) Risk Review Flag is a synthetic educational label.
FinTrust Data Dictionary	Field definitions, data types, and business/modelling use for both tables	Used to understand the meaning and intended use of variables during analysis, and the intentional missingness before profiling.	Some analytical meaning may require further exploration of the actual data.

3. Track Objectives

The Data Analytics track will pursue the following objectives during the project:

1. Profile and validate the quality & structure of the customer and transaction datasets.
2. Develop analytical questions, and KPIs that align with FinTrust's business problem and available data.
3. Analyze customer, transaction, channel, location, transaction-status, and risk-review patterns to identify meaningful business insights.

4. Design a dashboard wireframe that surfaces these KPIs, trends, comparisons, and the key risk patterns found.
5. Produce evidence-based findings that can support business decision-making and a realistic plan for building the full dashboard across Weeks 2–4.

4. Success Criteria

- Every analytical finding is backed by an actual computed statistic from the data, not assumption.
- Every proposed KPI can be calculated directly from fields that exist in the two datasets.
- The dashboard wireframe reflects patterns already validated in Week 1, not speculative visuals.
- The Week 4 dashboard reproduces the same baseline numbers established this week (success rate, risk rate, dormancy rate, etc.), confirming consistency.
- All data-cleaning and analysis decisions are documented well enough for another analyst to reproduce them.

5. Data Understanding / Profiling Report

Dataset Overview:

Two primary datasets were provided for the Data Analytics track: Customer Dataset and a Transaction Dataset.

Dataset	Number of Records	Number of Columns	Primary Purpose
Customer Data	1,500	12	Understanding customer characteristics and behaviour
Transaction Data	12,000	12	Understanding transaction activity and performance

The two datasets can be connected using Customer ID, which serves as the common identifier between customer and transaction records.

The relationship is one-to-many because one customer can have multiple transaction records.

5.1 Customer Dataset Structure

[Agu Precious FinTrust Customer Data clean.xlsx](#)

- Records: 1,500, 12 Columns
- No missing values, no duplicate Customer ID values.
- Numerical fields: Age (18–65), Tenure Months (1–96), Digital Engagement Score (16.7–100).
- Categorical fields: Gender, City (8 cities, Lagos largest at 488), Customer Segment, Account Type, Monthly Income Band, Preferred Channel, Account Status (1,367 Active / 107 Dormant / 26 Restricted).

5.2 Transaction Dataset Structure

[Agu Precious FinTrust Transaction Data clean.xlsx](#)

- Records: 12,000 transactions, 11 Columns
- Customer ID is a clean foreign key: all 1,500 customers appear in transactions, zero orphaned records
- Date/time: Transaction Date Time spans 1 Jan – 31 Mar 2026, all values parse correctly
- Numerical: Amount NGN ranges ₦100 to ₦693,454

- Categorical: Transaction Type, Channel, Device Type, Location, International Transaction (480 of 12,000 are international), Transaction Status (10,856 Successful / 630 Failed / 326 Reversed / 188 Pending), Risk Review Flag (2,352 Yes / 9,648 No)

5.3 Relationship between datasets

Customer ID links the two tables one-to-many (one customer to many transactions). The join is clean with no orphaned transaction records and no customers missing from the transaction table.

5.4 Missing values and data-quality issues

- 96 transaction rows are missing Device Type, 96 are missing Location, 4 rows are missing both, 92 are missing only one field. This matches the intentional missingness the data dictionary flagged.
- No missing values, duplicate keys, or orphaned records elsewhere in either table.
- Gender includes a 'Prefer not to say' category (51 customers), worth accounting for in any demographic breakdown.

6. Analytical Questions

1. Does Digital Engagement Score differ meaningfully between Active, Dormant, and Restricted customers?
2. Do different Customer Segment groups show different transaction frequency or average value?
3. Which Transaction Type is most common by Channel?
4. How does average Amount NGN vary by Monthly Income Band?
5. Is USSD used disproportionately in certain cities compared to Mobile App?
6. What share of transactions per channel end in Failed or Reversed status?
7. Is Risk Review Flag = Yes more common for certain Transaction Types? (Validated - see finding below)
8. Are international transactions flagged for risk review at a higher rate than domestic ones? (Validated - see finding below)

Validated findings (Q7 and Q8)

Risk flag rate by Transaction Type:

Transaction Type	Flagged %	Total Transactions
Transfer	28.5%	3,549
Cash Withdrawal	25.3%	1,430
Deposit	16.2%	1,328
Airtime/Data	15.2%	1,185
Card Purchase	13.0%	3,033
Bill Payment	12.8%	1,475

Risk flag rate — International vs Domestic:

Transaction Origin	Flagged %	Total Transactions
Domestic	18.9%	11,520
International	36.9%	480

The overall baseline flag rate across all transactions is 19.6%, confirming both patterns sit meaningfully above average.

7. KPI Definition Table

S/N	KPI	Definition	Why It Matters	Required Data	Baseline Value
1	Transaction Success Rate	Percentage of transactions Successful	Core operational health metric	Transaction Status	90.5%
2	Risk Flag Rate	Percentage of transactions flagged for review	Baseline for risk exposure monitoring. Provides a high-level view of risk-review activity	Risk Review Flag	19.6%
3	Average Transaction Value	Median amount (NGN) per transaction. $\frac{\text{Total transaction value}}{\text{total transaction volume}}$	Provides an indication of the average monetary value per transaction	Amount NGN, Transaction ID	₦10,306
4	Channel Adoption Rate	% of transactions per Channel	Helps understand channel usage and activity	Channel, Transaction ID	Mobile App 42.5%
5	Engagement by Status	Mean Digital Engagement Score by Account Status	Flags disengagement before dormancy	Digital Engagement Score, Account Status	Active 68.3 / Dormant 64.0
6	Dormancy Rate	% of customers Dormant or Restricted	Direct attrition-risk measure	Account Status	8.9%
7	International Risk Ratio	Risk flag rate: International vs Domestic	Highlights disproportionate international risk	International Transaction, Risk Review Flag	36.9% vs 18.9%
8	Failed/Reversed Rate by Channel	% of transactions failed or reversed, per channel	Identifies the weakest channel for reliability. Helps monitor unsuccessful transaction activity	Channel, Transaction Status	USSD 8.7% (highest)

8. Dashboard Wireframe

8.1 Proposed Dashboard Title

FinTrust Business & Transaction Intelligence Dashboard

8.2 Dashboard Objective

The dashboard will provide a consolidated view of FinTrust's customer and transaction activity. It will allow users to monitor key performance indicators, understand transaction and channel behaviour, examine customer and geographic differences, and explore risk-review patterns.

8.3 Proposed Design

- **Filters**

Date | Transaction Type | Channel | Status | Customer Segment | Account Type | Location | Risk Flag

- **KPI cards (top row)**

Total Transactions | Total Value | Success Rate | Average Value | Risk-Review Rate

- **Transaction performance**

Transaction Volume Trend | Transaction Value Trend

- **Transaction & channel analysis**

Transaction Volume by Type | Transaction Volume by Channel

- **Customer & status analysis**

Transaction Status Distribution | Transaction Activity by Customer Segment

- **Geographic & risk analysis**

Transaction Activity by Location | Risk-Review Patterns

8.5 Charts

- Transaction Volume by Channel
- Transaction Status Distribution
- Transaction Status Across Channels

This section will help users understand how customers use different channels and how transaction outcomes vary across those channels.

8.6 Major insights the dashboard is expected to communicate

- Transaction trends over time.
- Major transaction types by volume and value.
- Frequently used transaction channels.
- Transaction outcome patterns.
- Differences in activity across customer segments.
- Differences in activity across account types.
- Geographic differences in transaction activity.
- Associations between digital engagement and transaction behaviour.
- Concentration of risk-review flags across selected transaction dimensions.

Actual findings will be determined after the analysis is completed and will not be assumed at the dashboard-planning stage.

9. Initial Analysis Plan (Weeks 2–4)

9.1 Analysis Objective

The objective of the analysis is to transform the available customer and transaction data into meaningful business insights that support FinTrust's understanding of customer behaviour, transaction activity, channel usage, transaction outcomes, and risk-review patterns.

Week 2 — Data Preparation & Exploration

- Prepare and validate the datasets.
- Join customer and transaction data using Customer ID.
- Conduct exploratory data analysis.
- Calculate initial KPIs.
- Begin developing analytical visuals.
- Identify initial patterns that require deeper investigation.
- Resolve the 96 rows with missing Device Type/Location (impute, flag, or exclude)
- Cross-tabulate Risk Review Flag by City and Customer Segment
- Begin cross-track collaboration — check overlap with Data Science's Risk Review Flag modelling work.

Week 3 — Deeper Analysis & Dashboard Development

- Conduct deeper customer and transaction analysis.
- Analyze channel and transaction-status performance.
- Examine geographic patterns
- Build the working Power BI (or Excel) dashboard matching the Week 1 wireframe
- Add a time-series view using Transaction Date, Time (Jan–Mar 2026 trend)
- Add drill-through from KPI cards to transaction-level detail
- Validate dashboard KPI figures against this week's baseline numbers.

Week 4 — Finalization & Communication

- Polish dashboard visuals based on feedback
- Write final business-intelligence summary with recommended actions for FinTrust management
- Document key insights and limitations.
- Prepare the final analytical output.
- Coordinate with Project Management and other tracks for final integrated presentation.

10. Relevant Risks and Future Dependencies

Risks

- Data-quality decisions made in Week 2 (handling the 96 missing rows) could shift downstream KPI numbers.
- Tight timeline for a full dashboard build within a single week (Week 3).
- Some KPIs show only mild variation across categories, risk of overstating weak signals if not handled carefully.

Future Dependencies (planning only)

- The Data Science track may later require transaction-level insights or feature ideas from this analysis.
- The Project Management track will need this track's Week 2–4 timelines for the overall Work Breakdown Structure.
- The Generative AI track's escalation scenarios may reference risk categories identified here (e.g. Transfer/international risk).